Categorizing ODE

- Order: based on degree of derivative (i.e. $y'' \rightarrow 2^{nd}$ Order)
- homogeny: homogeneous $\rightarrow$ if ODE = 0 and/or all terms contain dependent variable or its derivative (ex: $y' = y + \sin(y)\, y''$)
  Non-homogeneous $\rightarrow$ terms in ODE involve the indepent variable or Constants w/o dependent Variable present (ex: $y(x) = 2y(x) + 5$ or $y'(x) = x + y$)
- Linear: all terms are linear w/respect to dependent variable

Analytical Methods for 1st order ODE general solutions

- Direct Integration (ex: $y'(x) = f(x) \Rightarrow y(x) = \int f(x)dx + c$)
- Seperation of Variables (ex: $y'(x) = g(x)\, h(y) \Rightarrow \frac{1}{h(y)} dy = g(x) dx$)
- Exact: $M(x,y) + N(x,y)\, y' = 0$ *general ODE form*
  * if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y} \Rightarrow$ exact else ODE is Not exact
  * if exact solve to find $\psi(x,y) = C$ $\left| \begin{array}{l} \frac{\partial \psi}{\partial x} = M(x,y) \\ \frac{\partial \psi}{\partial y} = N(x,y) \end{array} \right.$
  * integrate partials $\psi = \int M dx = \int M dy$
- Integration Factor: Applied if general 1st order ODE takes the form $y' + p(x)y = g(x)$ | $y(x) = e^{-\int p(x)dx}\left(\int g(x) e^{\int p(x)dx} dx + c\right)$

Analytical Methods for 2nd order ODE general solutions

- Direct Integration (ex: $y''(x) = f(x) \Rightarrow y(x) = \int\left(\int f(x)dx + c_1\right)dx + c_2$)
  Must be of form $y''(x) = f(x)$
- U Substitution/Order of reduction (of form $y''(x) = f(x, y')$)
  ex: $y'' + y' + y = 0 \quad u = y' \quad u' = y''$
  *Note for 3rd order or higher $\Rightarrow y''' + y'' + y' + y_0 = 0 \Rightarrow \{u'\} = [A]\{u\}$
  $y = y_1 \quad u_1 = y' = y_2 \quad u_2' = y'' = y_3 \quad u_3' = y''' = -y_3 - y_2 - y_1$
- Linear 2nd order ODE's: $y'' + p(x)y' + q(x)y = g(t)$ *if $g(t) = 0$ homogeneous
  $\Rightarrow$ 2 linear inpendent solutions (y_1, y_2)
  * denpending on $p(x)$ and $q(x)$ 2 diffrent Approaches
    - Constant Coefficient or Euler-Cauchy eqn.

Constant Coefficients

$y'' + ay' + by = 0$ Solutions have form $y = Ce^{rt}$

- use substitution $y = Ce^{rt}$, $y' = rCe^{rt}$, $y'' = r^2 Ce^{rt}$ into ODE divide out Ce^{rt} to get quadratic $r^2 + ar + b = 0$
- Solve quadric for (r_1, r_2)

root types	General equations
real/unequal roots $r_1 \neq r_2$	$y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$
real/equal roots $r_1 = r_2$	$y(t) = C_1 e^{r_1 t} + C_2 e^{r_1 t}$
Zero Root $r_1, r_2 = 0$	$y(t) = C_1 e^{r_1 t} + C_2$
Imaginary/Complex $r_{1,2} = r_r + r_i i$	$y(t) = C_1 e^{r_r t} e^{r_i t i} + C_2 e^{r_r t} e^{r_i t i}$ $y(t) = C_1 e^{r_r t}\sin(r_i t i) + C_2 e^{r_r t}\cos(r_i t i)$

Euler-Cauchy

- equation is of the form: $x^2y''+axy'+by=0$ and solutions take the form $y=x^m$ solve for y' and y'' to get characteristic equ. takes form $m^2+(a-1)m+b=0$
- **Real Roots:** $y(x)=C_1x^{m_1}+C_2x^{m_2}$
- **Repeated Roots:** $y(x)=C_1x^m+C_2x^m\ln(x)$
- **Imaginary/Complex Roots:** $m=\alpha\pm\beta i$ and $y(x)=x^{\alpha}(C_1\cos(\beta\ln(x))+C_2\sin(\beta\ln(x)))$

Continuous $F(t)$: Undetermined Coefficient

- Meant for Nonhomogeneous ODE's: $y''+ay'+by=F(t)$
  - Solve for Complementary eq. (i.e ODE $=0$, $y_h(t)$)
  - Solve for particular solution (i.e $y_p(t)$)
    - Using $y_p(t)$ based on $F(t)$ solve for y_p', y_p''
    - Substitute y_p, y_p', y_p'' into ODE and solve for y_p
  - general solution $y(t)=y_h(t)+y_p(t)$

If ODE is linear and System has form

$\{\dot{u}\}=[A]\{u\}$ and is 3rd order or larger

Solutions = eigen values of $[A]$

- $\text{eig}[A]=\det|A-\lambda I|$ $\quad I=\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$ identity matrix

ex:) $\text{eig}[A]=\det\left|\begin{bmatrix}0&1&0\\0&1&0\\-c&-b&-a\end{bmatrix}-\begin{bmatrix}\lambda&0&0\\0&\lambda&0\\0&0&\lambda\end{bmatrix}\right|$

$$=\begin{vmatrix}-\lambda&1&0\\0&-\lambda&1\\-c&-b&(a-\lambda)\end{vmatrix}\Rightarrow -\lambda\begin{bmatrix}-\lambda&1\\-b&-a-\lambda\end{bmatrix}-1\begin{bmatrix}0&1\\-c&-a-\lambda\end{bmatrix}+0\begin{bmatrix}0&-\lambda\\-c&-b\end{bmatrix}$$

$=-\lambda(-\lambda(a-\lambda)-1b)-(0-(-c))+0$

$=\lambda^2a\lambda-\lambda^3-b-c=0$ Solve for $\lambda_1,\lambda_2,\lambda_3$

general solution $y(t)=\sum_{n=1}^{3}C_ne^{\lambda_nt}$

Jacobian (Non linear systems)

- use Order Reduction to find $\{\dot{u}\}=[A]\{u\}$

$$J=\begin{bmatrix}\frac{\partial f}{\partial y_1}&\frac{\partial f}{\partial y_2}&\frac{\partial f}{\partial y_3}\\\frac{\partial g}{\partial y_1}&\frac{\partial g}{\partial y_2}&\frac{\partial g}{\partial y_3}\\\frac{\partial h}{\partial y_1}&\frac{\partial h}{\partial y_2}&\frac{\partial h}{\partial y_3}\end{bmatrix}$$

- evaluate J at local equilibria
  - Set dependent derivatives $=0$
  - Solve for dependent variables
- eig $\boxed{J(y_1,y_2)}$ Solve for eigenValues, G.S.

$F(t)$	$y_p(t)$
e^{-at}	Ke^{-at}
At	K_0+K_1t
t^n	$K_0+K_1t+K_2t^2+\dots+K_nt^n$

$F(t)$	$y_p(t)$
C	K
$\cos(\omega t)$	$K_1\cos(\omega t)+K_2\sin(\omega t)$
$\sin(\omega t)$	$K_1\cos(\omega t)+K_2\sin(\omega t)$

For Non-linear ODEs:

Small Angle Aprox.

$\sin(\theta)=\theta$ $\quad\tan(\theta)=\theta$

$\cos(\theta)=1$